

A DBTL-based Bioinformatics Pipeline for Comparative Metabolic Pathway Annotation in Microalgae

Dunaliella salina vs *Nannochloropsis gaditana*

José Campos

University of Minho · School of Engineering

Supervisor: Óscar Dias

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Context and Motivation

Dunaliella salina FACHB435

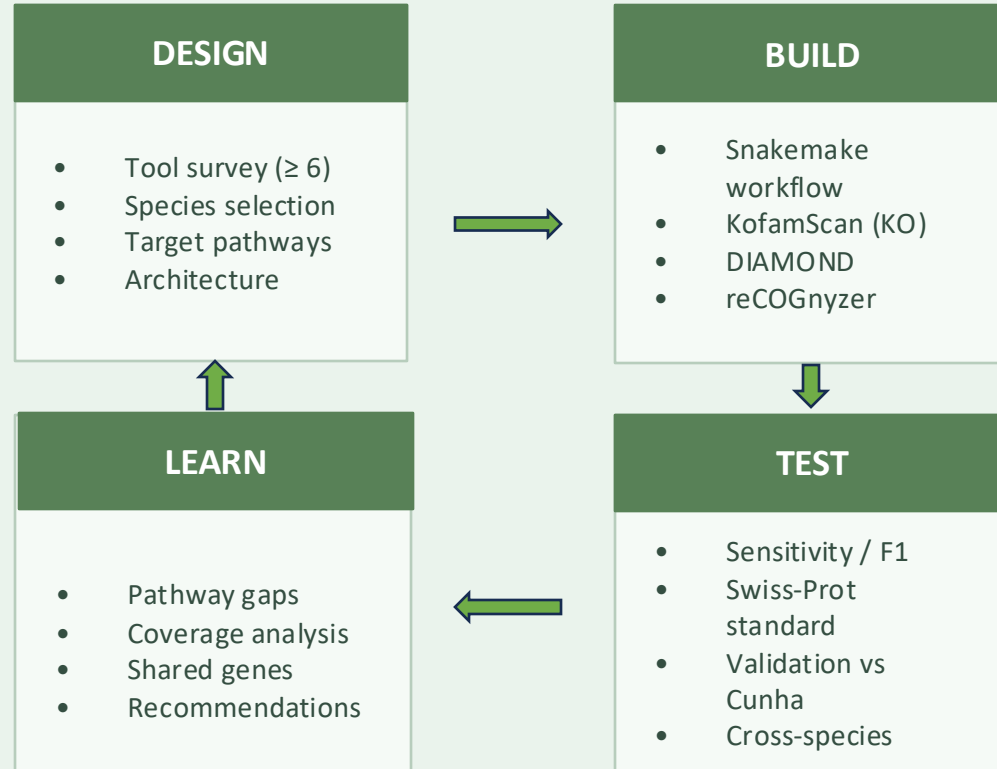
- Halotolerant green microalga
- β -carotene > 10% dry weight under salt stress
- No cell wall → easier extraction
- ~17,000 proteins (Chen et al., 2024)

Nannochloropsis gaditana B-31

- Marine eustigmatophyte
- High EPA (omega-3) — up to 38% of lipids
- Target for biofuels and nutraceuticals
- ~11,000 proteins (strain B-31)

Problem: Annotations are fragmented across heterogeneous databases (NCBI, KEGG, UniProt), produced with different tools and without a shared benchmark. GEM approaches require extensive manual curation. No automated, reproducible pipeline exists for cross-species comparison.

DBTL Framework



Adapted from Carbonell et al. (2018)

SMART Objectives

1

Tool survey

Evaluate ≥ 6 annotation and pathway-prediction tools. Build a comparison matrix: scope, I/O, licence, API, containerisation.

2

Reproducible pipeline

Integrate two tools (KofamScan + DIAMOND) into a Snakemake workflow. Deposit in a public GitHub repository.

3

Comparative annotation

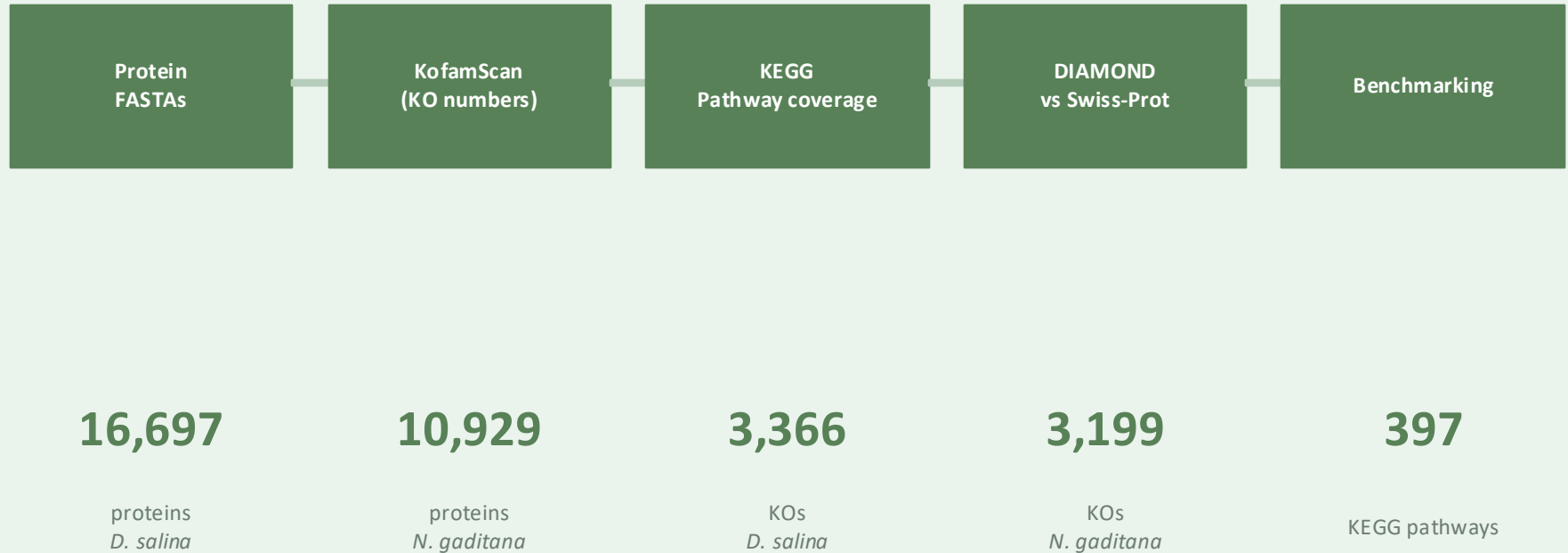
Compare the metabolic repertoires of both species, focusing on carotenoids (ko00906) and omega-3 (ko01040).

4

Benchmarking

Assess accuracy with sensitivity, specificity and F1 vs a Swiss-Prot gold standard. Validate against Cunha et al. (2025).

Pipeline Architecture



Tool Comparison

Tool	DBTL Phase	Function	Open-source
KofamScan	Design/Build	KO assignment (HMM)	Yes
DIAMOND	Test	Homology vs Swiss-Prot	Yes
reCOGnyzer	Build	COG annotation	Yes
KEGGCharter	Test/Learn	Pathway maps	Yes
Snakemake	Build	Workflow management	Yes

Two annotation tools (KofamScan + DIAMOND) integrated into the Snakemake workflow.

Results: Functional Annotation

KofamScan — KO assignment

- *D. salina*: 3,366 KOs (20.2% of 16,697 proteins)
- *N. gaditana*: 3,199 KOs (29.3% of 10,929 proteins)

DIAMOND vs Swiss-Prot — homology annotation

- *D. salina*: 42.2% annotated
- *N. gaditana*: 51.8% annotated

Cross-species comparison

1,709

Shared KOs

907

D. salina only

823

N. gaditana only

Results: KEGG Pathway Coverage

Coverage	D. salina	N. gaditana
Complete ($\geq 50\%$)	11	8
Partial (10–49%)	198	187
Absent ($< 10\%$)	188	202
Total evaluated	397	397

Target pathways

- Carotenoids (ko00906): Partial
- Omega-3 (ko01040): Partial
- Glycolysis (ko00010): Complete

Partial coverage is biologically expected: the KEGG reference KO sets include genes from bacteria, plants and fungi that microalgae do not possess.

Results: Desaturase Validation

Validated against Cunha et al. (2025) — PUFA / omega-3 enzymes

KO	Enzyme	<i>D. salina</i>	<i>N. gaditana</i>
K00507	$\Delta 9$ -desaturase	present	present
K10255	$\omega 6$ -desaturase	present	present
K10257	$\omega 3$ -desaturase	present	absent
K12418	$\Delta 4$ -desaturase	present	absent

Key finding: *D. salina* has the complete desaturase set. The gaps in *N. gaditana* B-31 are confirmed present in strain CCMP1894 — a strain annotation limitation, not a true biological absence.

Results: Benchmarking

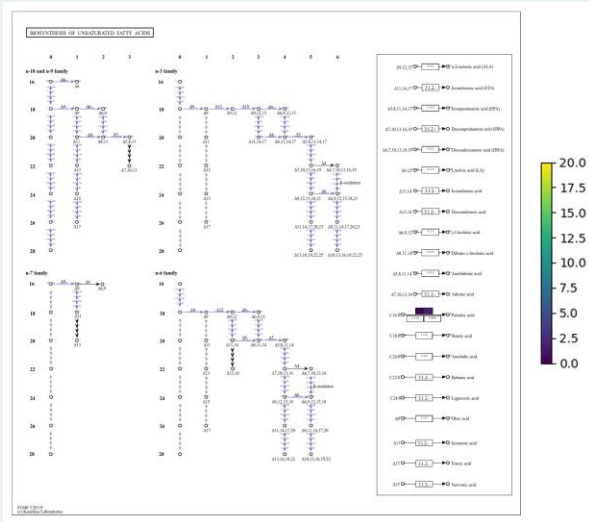
Gold standard: Swiss-Prot hits with $\geq 30\%$ identity and $\geq 50\%$ coverage

Metric	<i>D. salina</i>	<i>N. gaditana</i>
Sensitivity	0.53	0.61
Specificity	0.93	0.88
Precision	0.72	0.70
F1 score	0.61	0.65

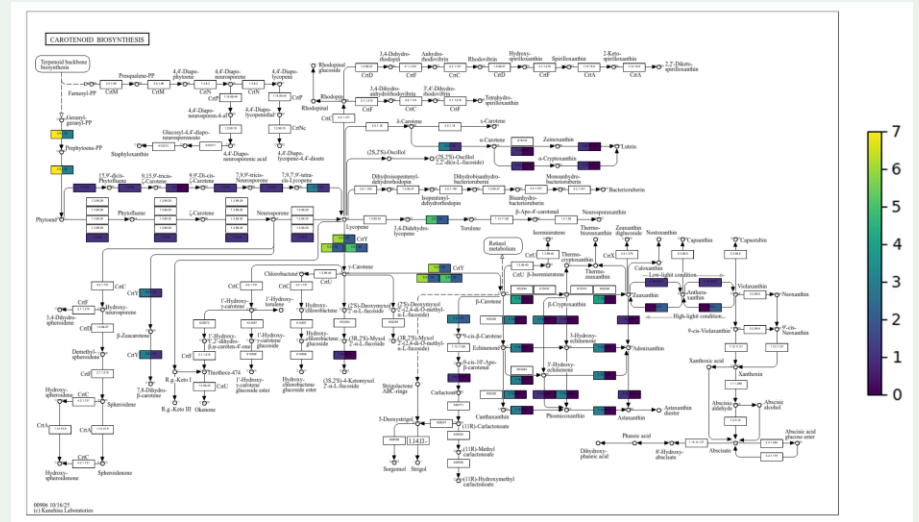
Complementary profiles:

N. gaditana shows higher sensitivity and F1, while *D. salina* has higher specificity and precision — both annotators perform comparably, with strengths in different metrics.

Results: Pathway Maps (KEGGCharter)



Comparison: differential Biosynthesis of unsaturated fatty acids



Comparison: differential Carotenoid biosynthesis

Conclusions and Future Work

Key conclusions

- Reproducible, open-source pipeline built in Snakemake
- *D. salina* has a complete PUFA desaturase set; *N. gaditana* B-31 shows strain-level gaps
- Partial coverage of carotenoid and omega-3 pathways is biologically expected
- Results validated against Cunha et al. (2025)

Future work

- Use strain CCMP1894 for *N. gaditana*
- Add transcriptomics and Docker containerization

AI Disclaimer

No AI tool was used for the elaboration of this presentation.